UNIVERSITY OF WOLLONGONG IN DUBAI

MMMB215 Project Submission 1

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Student declaration and acknowledgement (must be read by all students)

By submitting this assignment online, the submitting student declares on behalf of the team that:

1. All team members have read the subject outline for this subject, and this assessment item meets the requirements of the subject detailed therein.
2. This assessment is entirely our own work, except where we have included fully documented references to the work of others. The material contained in this assessment item has not previously been submitted for assessment.
3. Acknowledgement of source information is in accordance with the guidelines or referencing style specified in the subject outline.
4. All team members are aware of the late submission policy and penalty.
5. The submitting student undertakes to communicate all feedback with the other team members.

Submission 1: Model Winch for Foundry Engineering Design and Build Project

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Design Introduction

Task 1: Initial Design Concepts

Design Considerations:

The foremost objective of this project is to design and manufacture a model winch system for lifting crucibles in a metal foundry. The project will focus on developing a gear system that will optimize speed reduction on the input motor rpm, analyze shaft loading to ensure structural integrity and efficient power transmission, and predict the lifespan of the entire system. The project will have to consider the following design considerations:

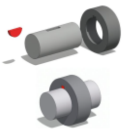
Table 1 Specification Summary Table

Specification	Requirement / Value
Lifted mass	1 kg
Lifting height	120 cm
Lifting time	10-20 s
Motor input power	2.1 W
Motor input speed	11,000 rpm
Gearbox configuration	3-stage reduction
Maximum gearbox housing size	200 mm x 200 mm
Gear backlash	0.1-0.2 mm
Pressure angle	20°
Shaft diameter at connection points	8 mm
Bearing type	608 bearing
Bearing dimensions	22 mm OD, 8 mm ID, 7 mm thickness
Input shaft protrusion length	50 mm
Output shaft protrusion length	100 mm
Budget limit	250 AED

Concept Designs & Morphological Chart:

The table below is a Morphological chart showcasing the ideas for various aspects of the project design:

Table 2 Morphological Chart

Design Function	Option 1	Option 2	Option 3	Option 4
Stage 1 Gear	Spur 	Helical 	Herringbone 	Bevel 
Stage 2 Gear	Spur 	Helical 	Herringbone 	Bevel 
Stage 3 Gear	Spur 	Helical 	Herringbone 	Bevel 
Shaft lock	Keyway 	Set Screw 	Splines 	Press fit 
Gear material	PLA filament 	Acrylic 	Wood 	
Shaft material	PLA filament 	Aluminum 		
Gearbox material	Acrylic 	MDF Wood sheets 		

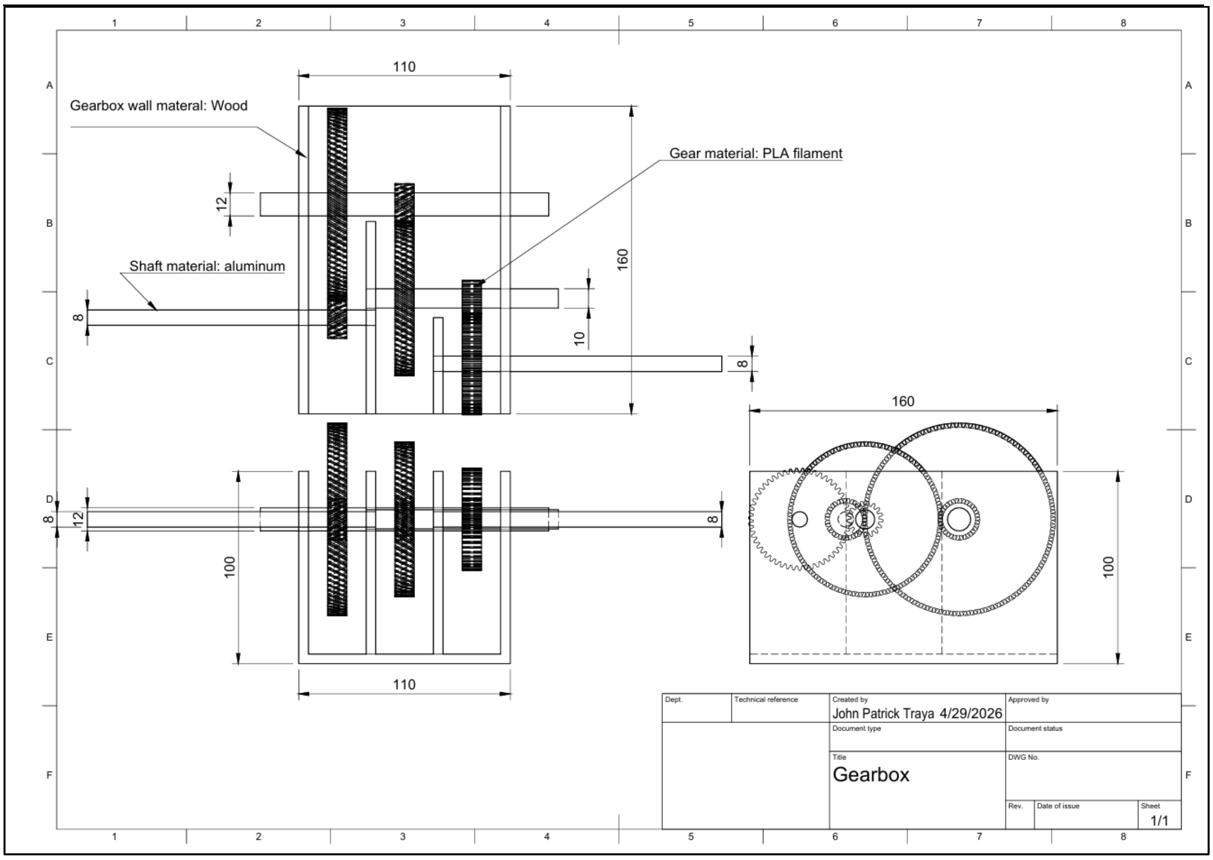


Figure 1 Concept Design 1

This design was based on a rough sketch drawn beforehand to get a visualization of the overall gearbox layout. This revised engineering drawing aimed to properly mark the dimensions and positions for the gears.

It features three stages of gear reduction with 5:1, 4:1, and 3:1 gear ratio respectively with the gears laid out in a horizontal manner. The shafts dealing with higher amounts of torque have wider shaft diameters of 12mm and 10mm. This ensures that they can handle the load sustained on them while minimizing the diameter enough to add the shaft lock mechanisms. The rough dimensions of the gears and the gearbox were based on calculations made for the center and pitch diameters of the gears to ensure accuracy towards finalizing the final model.

The materials used in the design were based on ideas from the morphological chart. For this concept, the gearbox walls are made of wood, providing structural integrity. The gears are made of PLA filament, and the shaft material is aluminum.

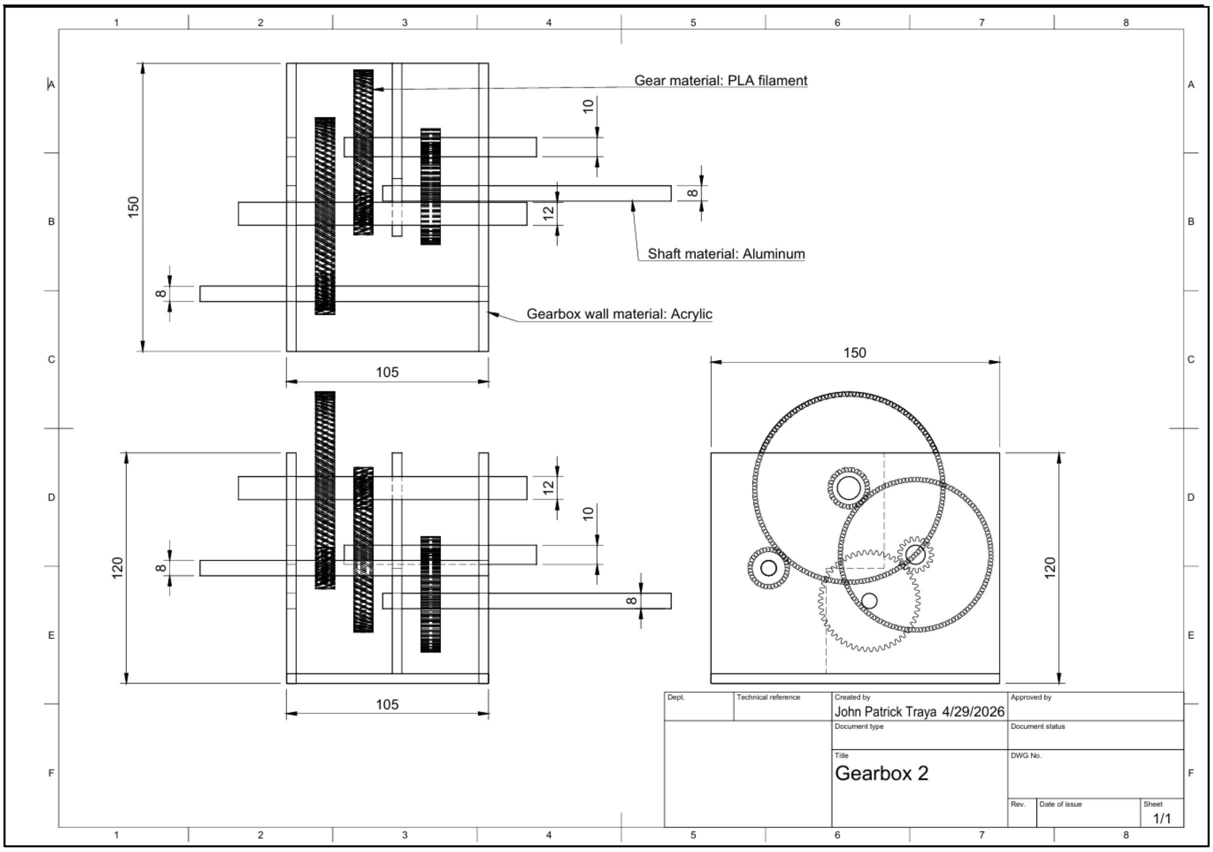


Figure 2 Concept Design 2

This design is like the previous design with the same gear dimensions and ratios with the difference being in their positions. The vertical configuration of the gears allows for a more compact design without affecting their performance, which is crucial given the large sizes of the gears being dealt with.

This model, just like the previous one, helps save space by utilizing supporting walls for the shafts. Unlike standard designs where the gears connect towards one direction which require a larger gearbox to hold, these designs place them in a more compact manner by changing the direction of connections to minimize as much space being taken up. The supporting walls still require extra materials to manufacture, but they still save materials used for making a larger gearbox.

The materials for this concept are the same as the last design, with the exception being the gearbox walls, which are made of Acrylic.

Design Reasoning & Pugh Matrix:

Table 3 Pugh Matrix Table

Evaluation Criteria		Concept 1		Concept 2	
	Priority- 5 is highest	Rating	Weighted	Rating	Weighted
Functionality	**5**	3	15	5	25
Strength	**5**	4	20	5	25
Speed	**3**	3	9	3	9
Control	**5**	4	20	4	20
Size	**4**	2	8	4	20
Manufacturability	**5**	5	25	4	20
Maintenance	**3**	3	9	2	6
Cost (higher means cheaper)	**4**	2	8	4	16
Complexity (higher means less complex)	**4**	3	12	2	8
Total Score			**126**		**149**

Both designs have their respective pros and cons but overall, the second concept design proved more useful for our case. The following are the main reasons for choosing this design:

- **Compactness**
  Due to the differing gear elevation, the added vertical distancing allows for more space for the shaft and gears to sit in, unlike the first design which has its gears all under one horizontal elevation. This provides the added benefit of reducing gearbox dimensions while not affecting the performance of the gears.

- **Lower cost**
  As a result of the compactness, the amount of materials needed to build the gearbox will be lower, resulting in a lower manufacturing cost.
- **Structural integrity**
  Due to the nature of the design, the large gear on the first stage blocked the wall for the second stage shaft to attach to. To fix this, an additional supporting wall needed to be placed for the shaft to attach to. Despite this, it brought the extra benefit of added structural integrity and strengthening of the gearbox composition.
- **No performance impact**
  The gear and shaft placement do not have any impact on the performance of the gearbox as a whole. The force and torque transfer will still be the same as long as the gears are connected to each other.
- **Material viability**
  For the gearbox walls, Acrylic is proven to be the better choice for this aspect for a variety of reasons. First, it allows for more precise alignment during manufacturing. The gears generally need exact distances. Acrylic is less likely to warp than wood, ensuring the gears stay meshed correctly over time. Second, it is a preferred material for laser cutting, producing perfectly square, polished edges that are ready for assembly. Lastly, for this project, it would allow for the visibility of internal components. Allowing for the identification of any problems that may occur.

In addition, since the gears being used in the final design will be helical, 3D printing the gears will be the best choice to handle the manufacturing complexity of helical gears. Thus, the gears will be made from PLA filament. However, the chosen design comes at the cost of a slightly more complex manufacturing and assembly procedure. But the benefits it has still outweigh these disadvantages.

Task 2: Gear/pinion Design Calculations

- **Gear Ratio:**

  To determine the required gear ratio, we must first determine the required output shaft RPM using some basic equations.

  Here is the initial FBD of the output shaft and the mass hanging from it:

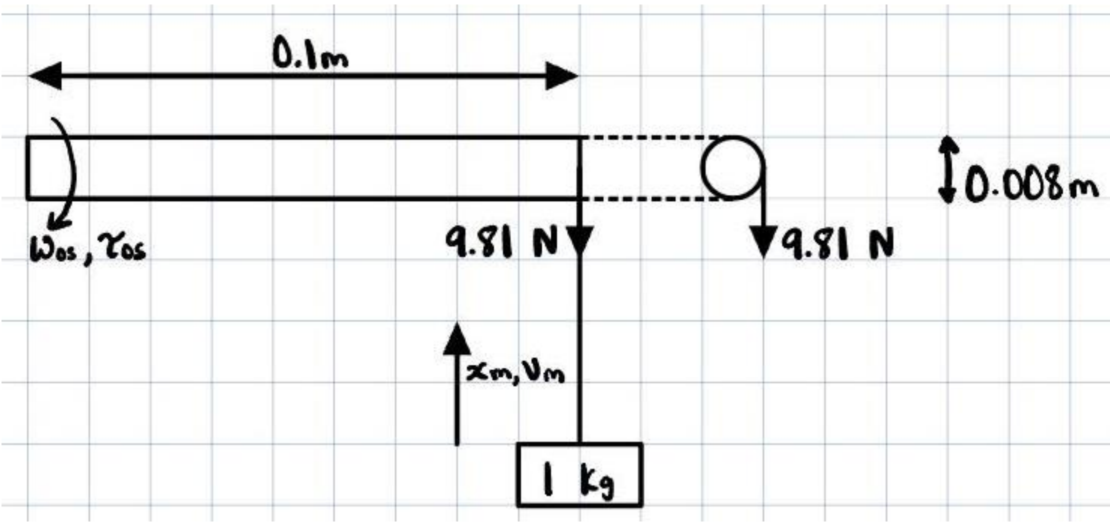


Figure 3 Initial FBD of Output Shaft

Where:

ω_{OS}	Angular velocity of the output shaft
τ_{OS}	Torque of the output shaft
x_m	Displacement of mass
v_m	Velocity of mass

To find ω_{OS}, we can use:

$$v_m = r\omega_{OS} = \frac{x_m}{t}$$

With the constraints being $x_m = 1.2\ m$ and $t = [10,20]\ s$. Which gives us the following result:

$$0.004\omega_{OS} = \frac{1.2}{(t = 15)} \rightarrow \omega_{OS} = 20\ rad/s \approx 191\ RPM$$

This means that our total gear ratio is $G = 11000/191 \approx 58$. Now, we can split this ratio into three stages: $G_1 = 5,\ G_2 = 4,\ \&\ G_3 = 3$.

Table 4 Gear Teeth for Each Stage

Stage Number	Pinion	Gear
Stage 1	17	85
Stage 2	17	68
Stage 3	17	51

Stage 1

For this stage, the chosen type of gear is helical, since the RPM reduces from 11000 to 2200, we cannot use spur gears as they would generate too much noise and vibrations at this speed.

Table 5 Stage 1 Gear Specifications

Normal module	m_n	1.5 mm
Normal pressure angle	ϕ_n	20 deg
Helix angle	ψ	30 deg
Teeth – Pinion	z_1	17
Teeth – Gear	z_2	85

Using the selected values, we can find the pitch circle diameter and addendum for both gears as well the centre distance, using the equations below:

$$a = \left(\frac{z_1 + z_2}{2}\right)\left(\frac{m_n}{\cos\psi}\right) \qquad d = \frac{z m_n}{\cos\psi} \qquad h_a = m_n$$

Table 6 Stage 1 Gear Parameters

Gear Parameter	Pinion	Gear
a	88.3346	
d	29.4449	147.2240
h_a	1.5	1.5

Using this information, we can create a 2D sketch of stage 1.

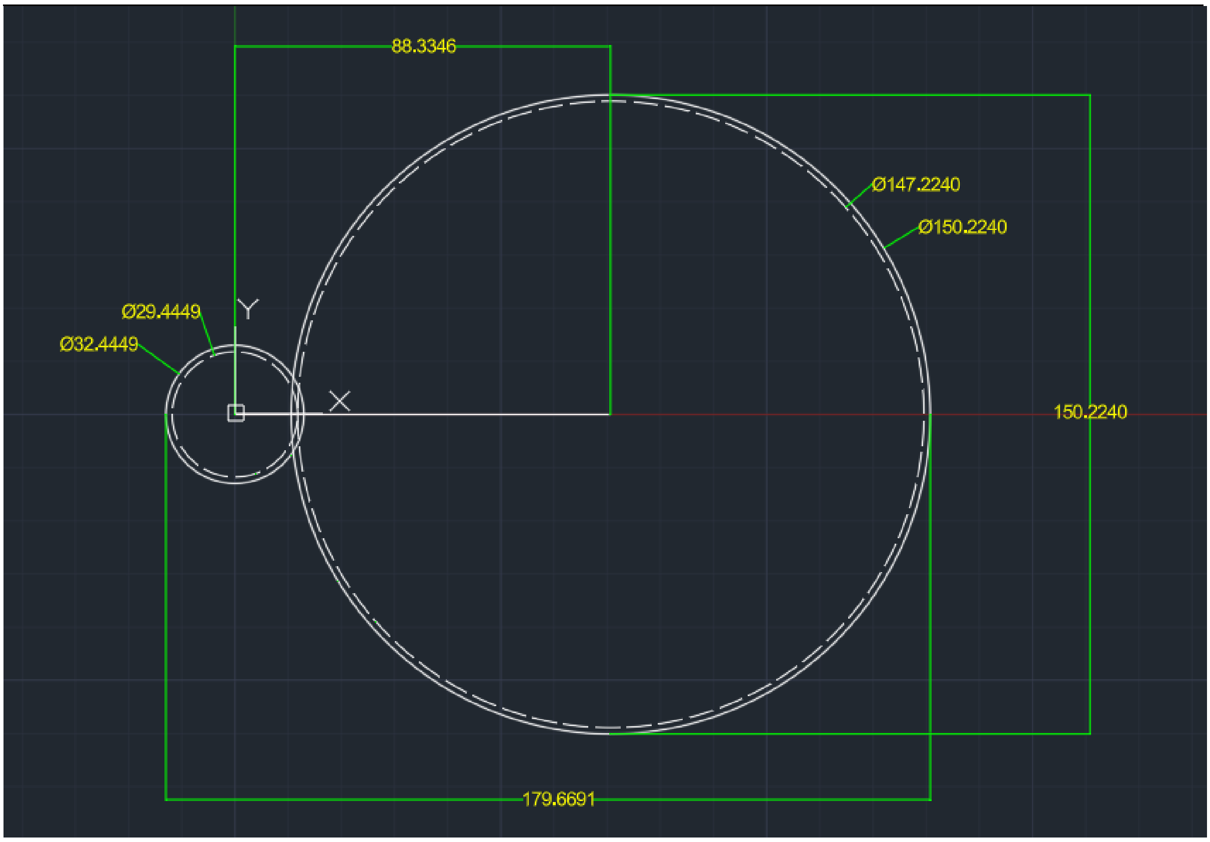


Figure 4 Stage 1 2D Sketch

Stage 2

For this stage, we will still use helical as the speed is still >1000 RPM.

Table 7 Stage 2 Gear Specifications

Normal module	m_n	1.5 mm
Normal pressure angle	ϕ_n	20 deg
Helix angle	ψ	30 deg
Teeth – Pinion	z_1	17
Teeth – Gear	z_2	68

Using the selected values, we can find the pitch circle diameter and addendum for both gears as well the centre distance, using the same equations above.

Table 8 Stage 2 Gear Parameters

Gear Parameter	Pinion	Gear
a	73.6122	
d	29.4449	117.7790
h_a	1.5	1.5

Using this information, we can create a 2D sketch of stage 2:

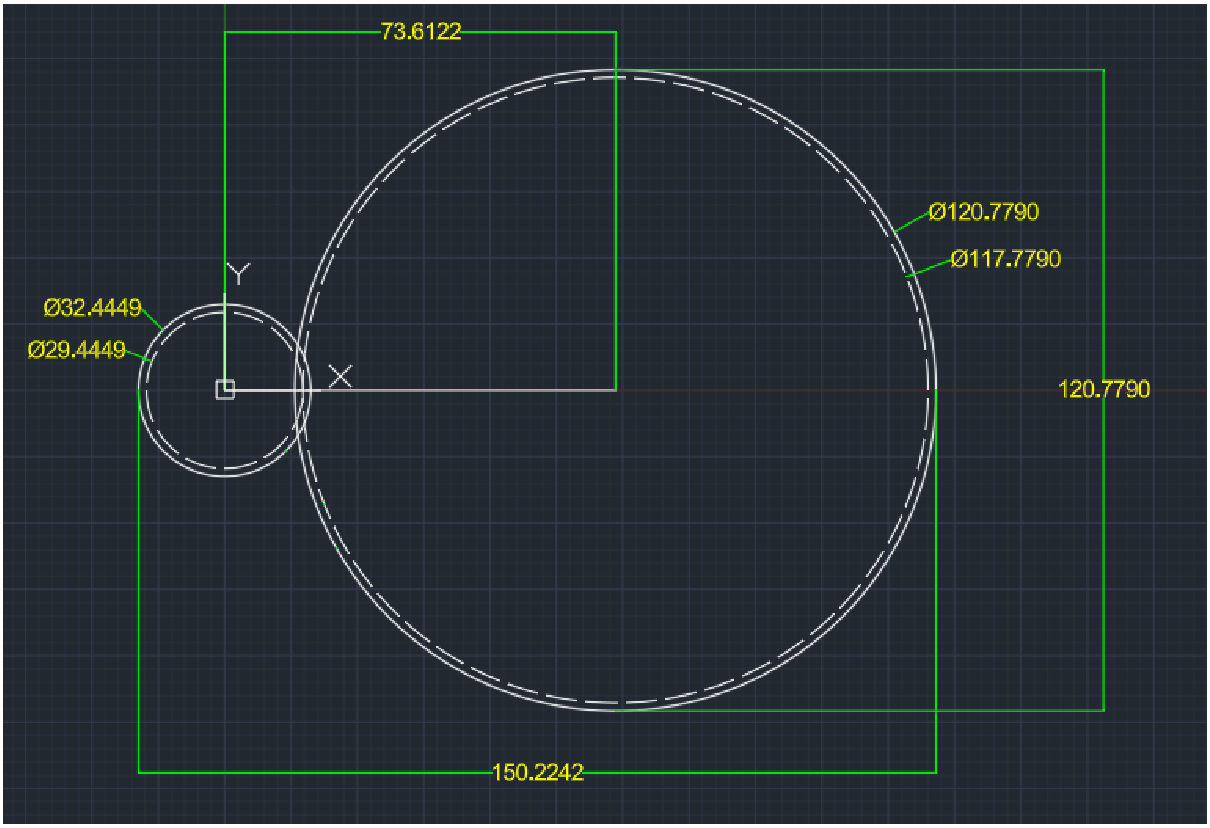


Figure 5 Stage 2 2D Sketch

Stage 3

For this stage we could go either spur or helical, since the speed is relatively low. Going with helical would be better as the rest of the gears are also helical, so it would be cheaper.

Table 9 Stage 3 Gear Specifications

Normal module	m_n	1.5 mm
Normal pressure angle	ϕ_n	20 deg
Helix angle	ψ	30 deg
Teeth – Pinion	z_1	17
Teeth – Gear	z_2	51

Using the selected values, we can find the pitch circle diameter and addendum for both gears as well the centre distance, using the same equations above.

Table 10 Stage 3 Gear Parameters

Gear Parameter	Pinion	Gear
a	58.8897	
d	29.4449	88.3346
h_a	1.5	1.5

Using this information, we can create a 2D sketch of stage 3.

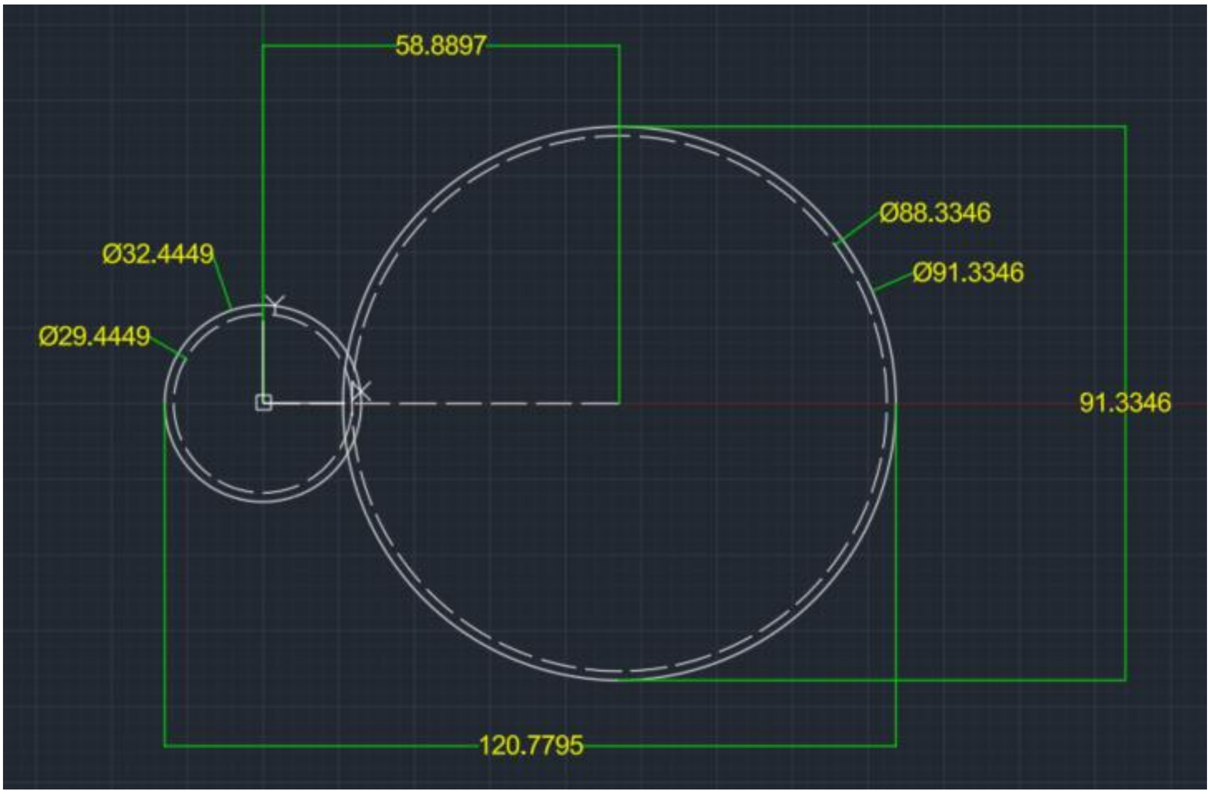


Figure 6 Stage 3 2D Sketch

➢ **Gearbox Dimensions:**

Now that we have the 2D sketches of the stages, we can predict the dimensions of the final gear box. The maximum width occupied by the gears is $\approx 180mm$. Which means the width of the gearbox must be around 195mm.

The maximum height occupied by the gears is $\approx 150mm$. This means that the height of the gearbox must be around 160mm.

As for the length, assuming each gear has a maximum thickness of 25mm, with some clearance and extra space, we could approximate a value around 105mm.

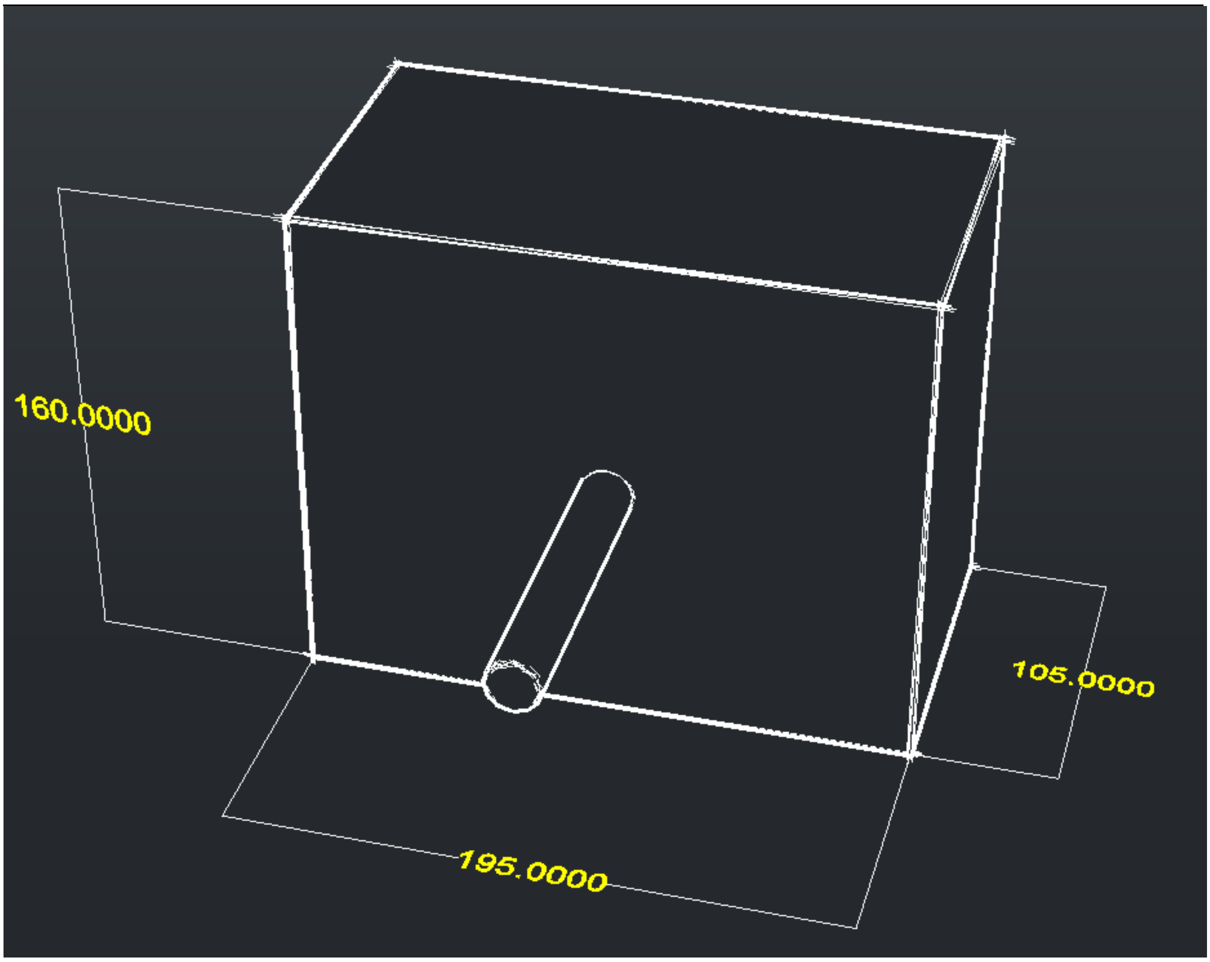


Figure 7 Gearbox Dimensions 1 2D Sketch

Task 3: Force Analysis

➢ **Torque and Angular Velocity of Shafts:**

a. TORQUE

STAGE 3 (Spur gears)

Output Shaft

Variables

d_{output} = 8 mm = 0.008 m, r_{output} = 0.004 m (Output shaft radius)

F_g = mg = 1 kg x 9.81 N/kg = 9.81 N (Force due to weight)

Calculations

$$\tau_{output} = F_g \times r_{output} = 9.81\ N \times 0.004\ m$$

$$\boldsymbol{\tau_{output} = 0.03924\ N.m}$$

Gear 3

Variables

$r_3 = d_3/2$ = 76.5/2 = 38.25 mm = 0.03825 m

r_{p3} = 25.2/2 = 12.75 mm = 0.01275 m (radius of pinion 3)

Calculations

1. Find W_{t3} (transmitted load on gear 3)

$$W_{t3} = \frac{\tau_{OS}}{r_3}$$

$$W_{t3} = \frac{0.03924\ Nm}{0.03825\ m} = 1.026\ N$$

2. Find torque on shaft B

$$\tau_B = W_{t3} \times r_{p3} = 1.026\ N \times 0.01275\ m$$

$$\boldsymbol{\tau_B = 0.012308\ N.m}$$

STAGE 2 (Helical Gears)

Gear & Pinion 2

Variables

$r_{g2} = d_{g2}/2 = 117.779/2 = 58.89$ mm $= 0.05889$ m (Gear 2 radius)

$\tau_B = 0.012308$ Nm

$r_{p2} = d_{p2}/2 = 29.4449/2 = 0.01472$ m (Pinion 2 radius)

Calculations

1. Find W_{t2} (transmitted load between stage 2 pinion and gear 2)

$$W_{t2} = \frac{\tau_B}{r_2} = \frac{0.012308\, Nm}{0.05889\, m}$$

$W_{t2} = 0.222$ N

2. Find torque on shaft A

$$\tau_A = W_{t2} \times r_{p2} = 0.222\, N \times 0.01472\, m$$

$\tau_A = 0.00327$ Nm

STAGE 1 (Helical Gears)

Variables

$\tau_A = 0.00327$ Nm

$r_{g1} = d_{g1}/2 = 147.224/2 = 73.612$ mm $= 0.07361$ m (Gear 1 radius)

$r_{p1} = d_{p1}/2 = 29.4449/2 = 0.01472$ m (Pinion 1 radius)

Calculations

1. Find W_{t1} (transmitted load between motor pinion and gear 1)

$$W_{t1} = \frac{\tau_A}{r_{g1}} = \frac{0.00327\, Nm}{0.07361\, m}$$

$W_{t1} = 0.0444$ N

2. Find motor input torque (τ_{in})

$\tau_{input} = W_{t1} \times r_{p1} = 0.0444\,N \times 0.01472\,m$

τ_{input} = 0.00065 Nm

b. ANGULAR VELOCITY

We use the gear ratios ($G_1 = 5$, $G_2 = 4$, $G_3 = 3$) to reduce the speed at each stage.

1 rad/s = 1 rpm x $\frac{2\pi}{60}$

- Motor speed ω_{in} = 11,000 rpm = **1,151.9 rad/s**
- Shaft A ω_A = 11,000/5 = 2,200 rpm = **230.38 rad/s**
- Shaft B ω_B = 22000/4 = 550 rpm = **57.6 rad/s**
- Output Shaft ω_{out} = 550/3 = 183.3 rpm = **19.2 rad/s**

Table 11 Result Summary Table for Torque and Angular Velocity

Stage	Torque (Nm)	Angular Velocity (rad/s)
1 (Spur)	0.00065	230.38
2 (Helical)	0.00327	57.6
3 (Helical)	0.012308	19.2

➢ **Pitch-line Velocity and Transmitted Load of Gears:**

c. TRANSMITTED LOAD OF GEARS

We find the axial and radial forces to understand what forces the gears are experiencing, hence figure out if the bearings will fail to handle such forces, as well as the shafts which will experience a bending moment because due to said forces.

$\psi = 30^{\circ}$ (Helix angle)

$\phi = 20^{\circ}$ (Pressure angle)

For HELICAL

$$W_r = W_t \times \frac{\tan(\phi_n)}{\cos(\psi)}$$

$$W_a = W_t \times \tan(\psi)$$

FOR SPUR

$$W_r = W_t \times \tan(\phi)$$

STAGE 1

<u>Radial force W_{t1}</u>

$$W_{r1} = 0.0444 \times \frac{\tan(20^{\circ})}{\cos(30^{\circ})}$$

W_{r1} = 0.0186 N

<u>Axial Force W_{a1}</u>

$$W_{a1} = 0.0444 \times \tan(20^{\circ})$$

W_{a1} = 0.0256 N

STAGE 2

<u>Radial force W_{r2}</u>

$$W_{r2} = 0.222 \times \frac{\tan(20^{\circ})}{\cos(30^{\circ})}$$

W_{r2} = 0.093 N

<u>Axial force W_{a2}</u>

$$W_{a2} = 0.222 \times \tan(30^{\circ})$$

W_{a2} = 0.128 N

STAGE 3 (SPUR GEARS)

Since spur gears have no helix angle, the formula is very straightforward:

$$W_{r3} = 1.026 \times \tan(20^{\circ})$$

W_{r3} = 0.373 N

d. PITCH LINE VELOCITY

The linear speed at the point where the gear teeth meet. For any two gears in mesh, the pitch-line velocity is the same for both the pinion and the gear.

$$V = \omega \times r$$

(Where ω is in rad/s and r is the pitch radius in meters)

Stage 1 $\omega_{in} = 1151.9$ rad/s, $r_{p1} = 0.01472$ m

Stage 2 $\omega_A = 230.4$ rad/s, $r_{p2} = 0.01472$ m

Stage 3 $\omega_B = 57.6$ rad/s, $r_{p3} = 0.01275$ m

- V_1 = **16.96 m/s**
- V_2 = **3.39 m/s**
- V3 = **0.73 m/s**

Table 12 Result Summary Table for Tangential and Radial Force and Pitch Line Velocity

Stage	Tangential W_t (N)	Radial W_r (N)	Pitch Line Velocity (m/s)
1 (Spur)	1.026	0.373	16.96
2 (Helical)	0.222	0.093	3.39
3 (Helical)	0.044	0.019	0.73

Checking for Lifting Velocity:

The pitch-line velocity of the output shaft surface (where the string is) should match the lifting speed needed for the 1.2 m height.

$$V_{lift} = \omega_{out} \times r_{shaft}$$

$$V_{lift} = 19.2 \times 0.004 = 0.0768\, m/s$$

Time to lift 1.2 m = $\frac{1.2}{0.0768}$ = **15.6 seconds,** this clears the 10 -20 s requirement.

Task 4: Input/output & Intermediate Shafts Design

Keys were chosen as the torque transmission element in this design. In mechanics, a key is defined as an element in a machine that is used to connect a transmission shaft to a rotating machine element. They are used in order to prevent relative rotation between the shaft and the attached component. Keys work by transmitting the torque caused by the shaft to the hub/rotating machine elements, which are gears in this case. To secure the shaft to the rotating element using a key, a keyway must be machined into either the shaft or the hub so as to accommodate the key. The creation of a keyway is usually the main drawback of using keys as a transmission element, since it creates a stress concentration that weakens the shaft.

There are many different types of keys, however this design will only use two: square keys and a Woodruff key. While using different key types would increase the manufacturing costs due to the additional complexity, the payoff in efficiency will outweigh the disadvantage posed.

Square and Woodruff keys are both types of sunk keys. A sunk key is a type of key that fits halfway into the keyway on the shaft and halfway into the keyway on the hub. Sunk keys transmit power through the key's shear resistance, which also prevents relative rotational motion between the shaft and the hub. Sunk keys are thus more appropriate for transmitting higher loads and torques.

Square Keys:

Usually fitted with a clearance at the top of the keyway in the hub, square keys transmit torque through the side faces that are embedded in the shaft. Since the shaft diameters in this design are all under 22mm, the keys used will have a square cross section.

There are many advantages to using square keys, mainly because they are very suitable for transmitting heavy loads or higher torques since they offer higher resistance to shear. Additionally, square keys are widely standardized and have low manufacturing costs. They are also easy to install and remove from a shaft. However, square keys are not suitable for high speeds due to created stress concentrations and the possibility of causing vibration or noise. Moreover, they usually require axial locking in order to prevent the hub from sliding back and forth along the shaft.

The advantages and disadvantages of using a square key are summarized in Table 13 below:

Table 13 Advantages and Disadvantages of Square Keys

Advantages	Disadvantages
• Easy Installation • Prevents relative motion • Transmits heavier torques and loads	• Not optimized for high speeds • Requires axial locking

Woodruff Keys:

Woodruff keys are semi-circular in shape with a flat upper surface. They are fitted into a similarly shaped keyway in the shaft. The deeper keyway works to prevent the key from slipping over the shaft. Their precise and more centered fit makes them ideal for high-speed applications, and thus a Woodruff key was chosen for the input shaft instead of a square key for the input shaft.

Woodruff keys are advantageous in various ways, for example, their shape allows for the key to be inserted deeply inside the shaft, which ensures easy assembly without the need for axial locking. This makes them ideal for high-speed applications since the depth provides stability and reduces rotational vibration. Furthermore, their shape also makes it easier to dislodge them in case they become stuck inside the shaft. Woodruff keys also improve concentricity between the shaft and the hub since it adjusts and aligns itself on the shaft. Finally, they do not allow for axial movement, which is ideal for this design.

Nonetheless, using a Woodruff key weakens the shaft due to the deeper keyway, which increases the stress concentration and reduces strength. This also means that are not suitable for higher loads and torques as they are often short in length. Their installation is also difficult, since the rounded bottom of the key can cause it to rotate or fall out during assembly.

The advantages and disadvantages of Woodruff keys are summarized in Table 14:

Table 14 Advantages and Disadvantages of Woodruff Keys

Advantages	Disadvantages
• Suitable for high-speed applications • Does not permit axial movement • Improves concentricity between shaft and hub • Easy to dislodge • Does not roll out of keyway • Does not require axial locking	• Not suitable for transmission of high torque • Difficult installation • Weakens shaft

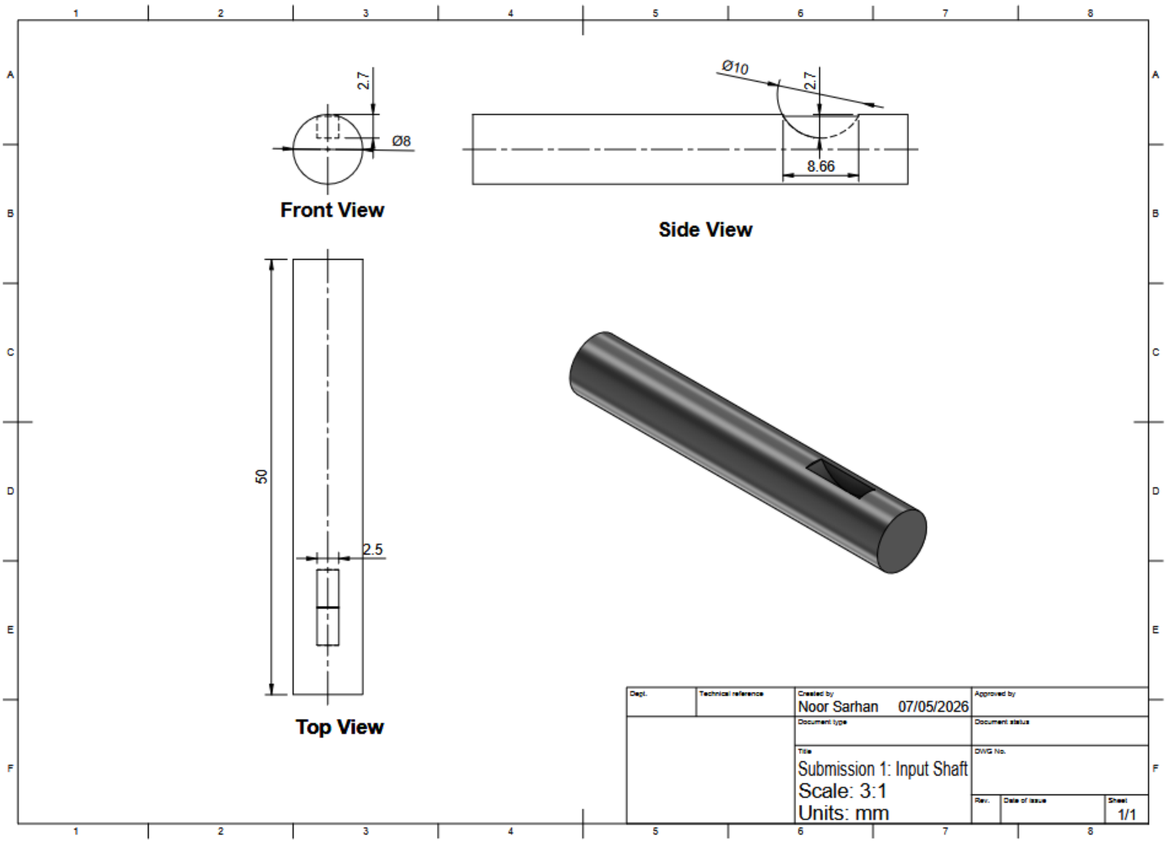


Figure 8 Input Shaft Design

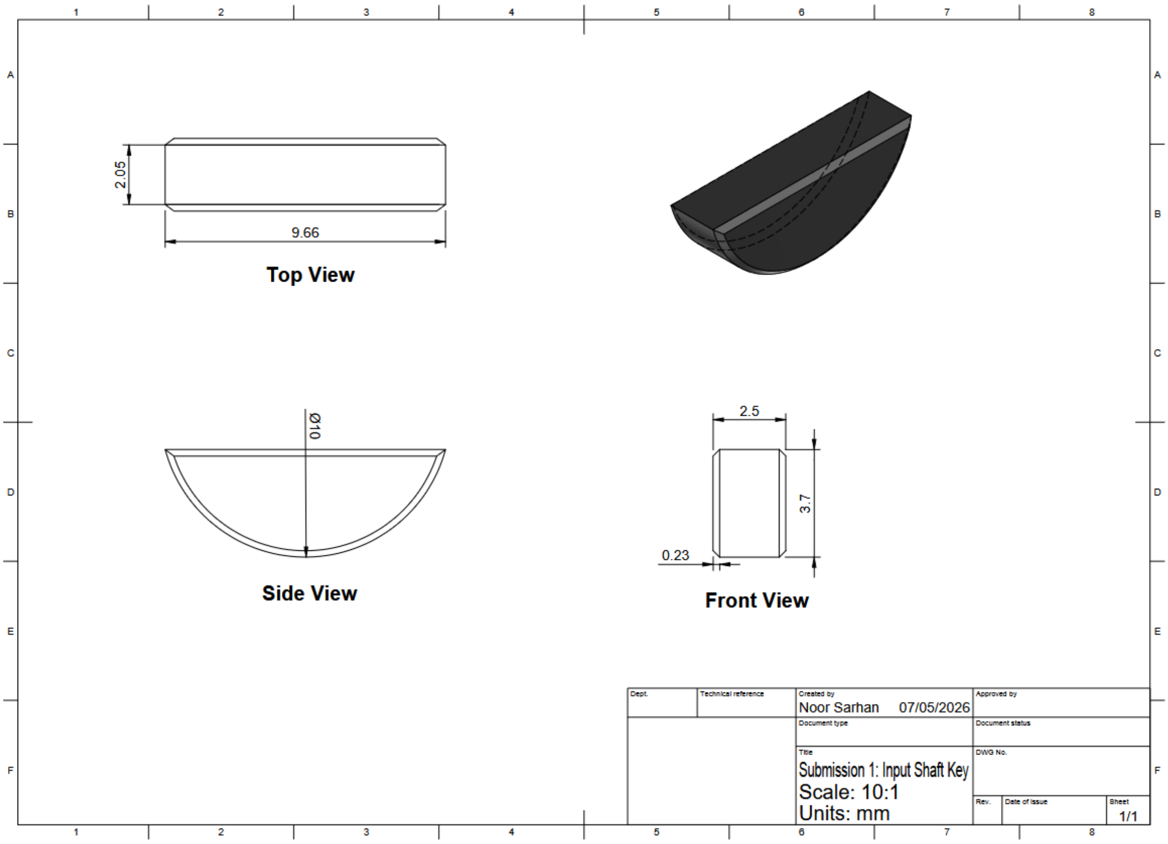


Figure 9 Input Shaft Woodruff Key Design

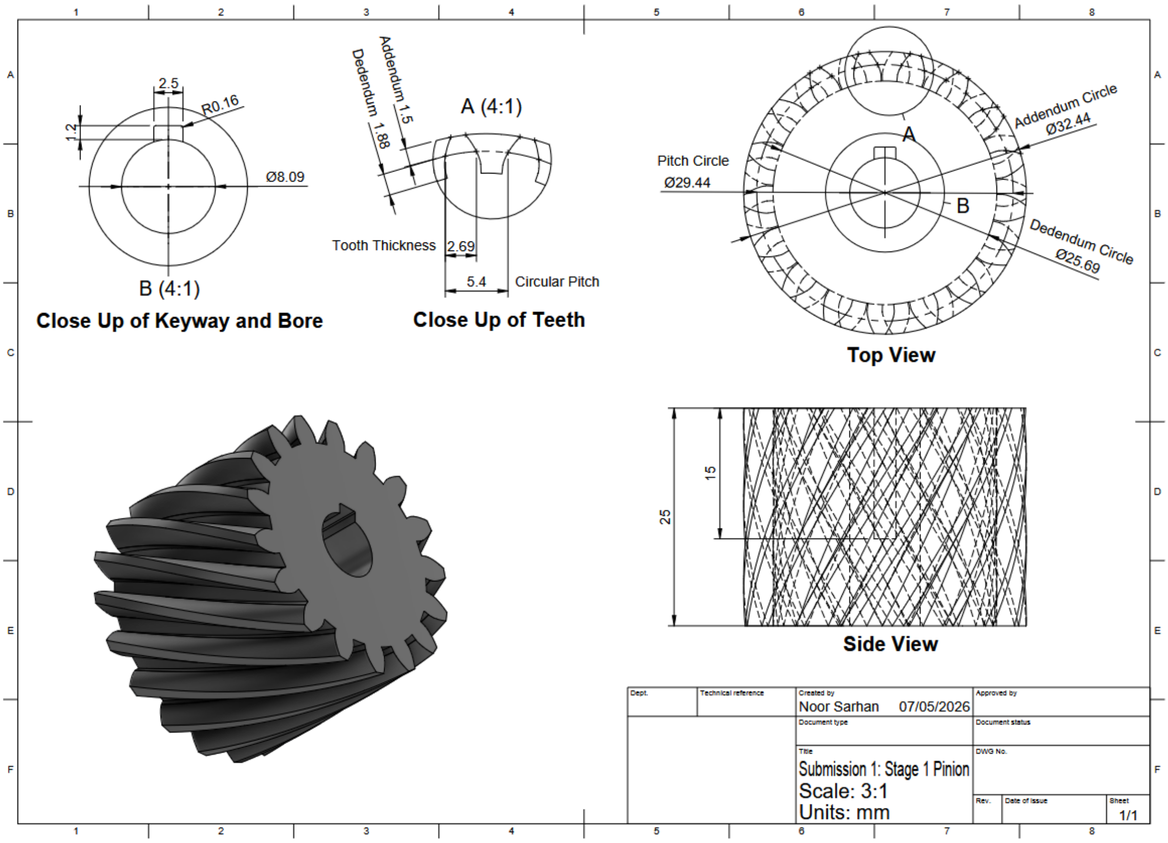


Figure 10 Stage 1 Pinion Gear

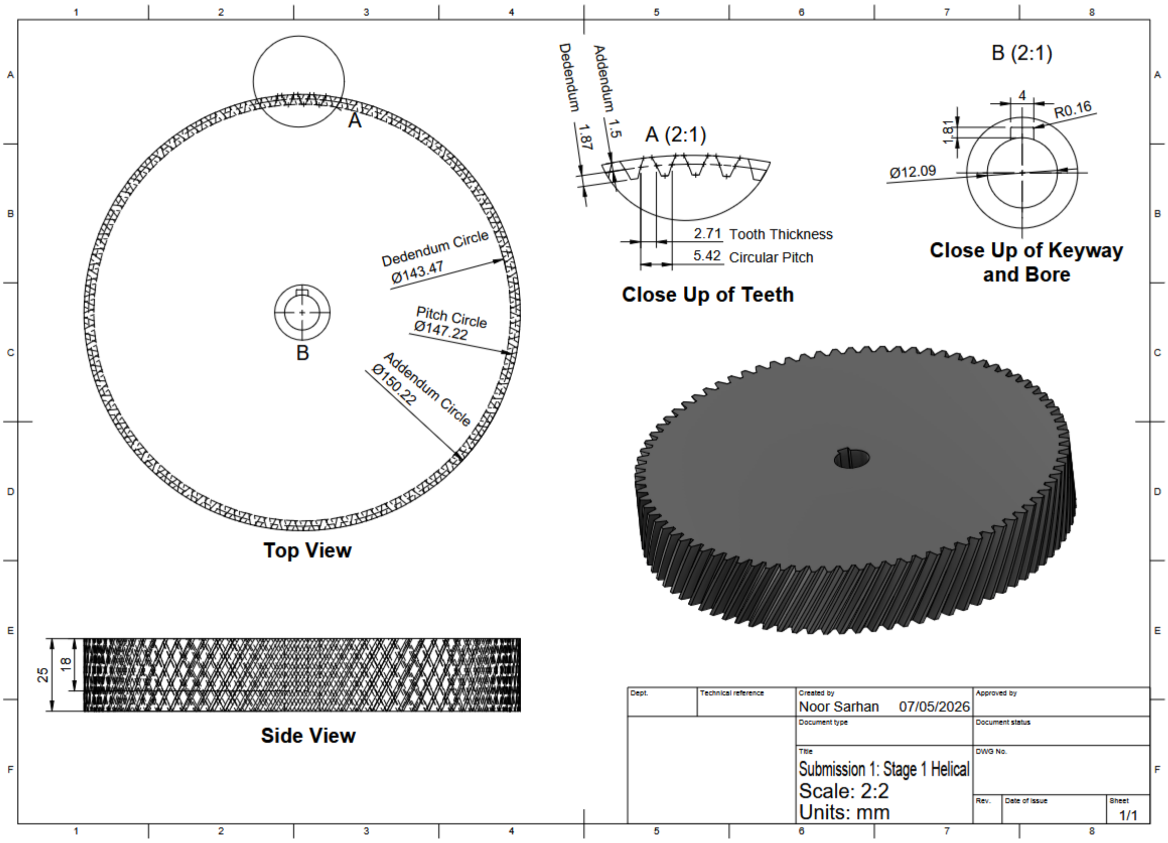


Figure 11 Stage 1 Helical Gear

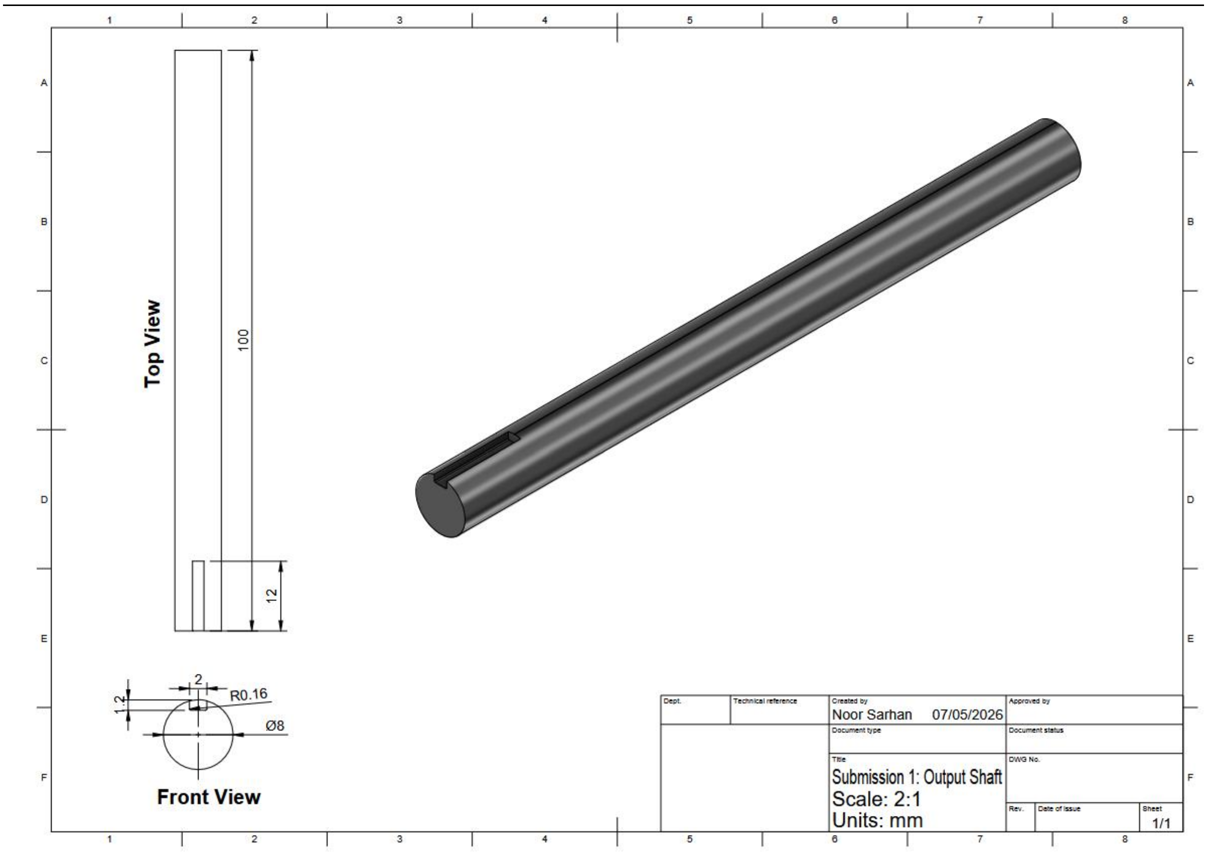


Figure 12 Output Shaft Design

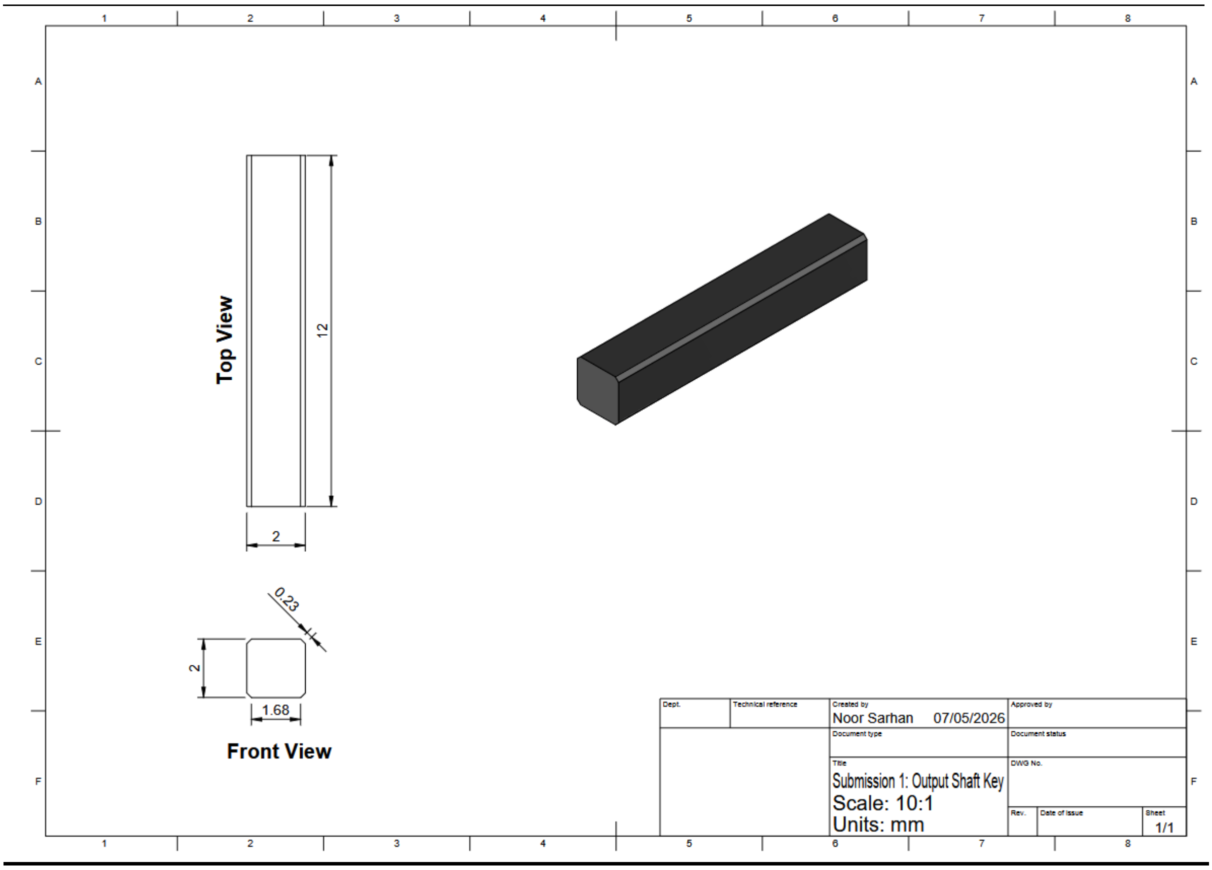


Figure 13 Output Shaft Square Key Design

Project Summary & Future Steps

In this project, a three-stage gearbox was developed for a winch system made to lift a 1 kg foundry crucible through 1.2 m within the required 10-20 seconds timeframe. Through concept generation, Pugh matrix evaluation, gear ratio calculations, force analysis and shaft/key selection, concept design 2 was chosen for several reasons: the compact structure, less cost, and easier manufacturability. The overall design has a 58:1 reduction ratio with helical gearing for smooth power transfer and aluminum shafts with Woodruff/square keys for torque transmission. Analytical calculations confirmed that the final design satisfies the speed, torque and lifting requirements while staying within the required gearbox dimensions.

For future steps, we will do the following:

1. CAD Modeling and Assembly

The 3D CAD models of all the gears, shafts, bearings, housing and keys will be finalized. As well as adding exact tolerances for backlash.

2. Material Validation

An evaluation will be conducted to decide whether PLA can withstand repeated loading, wear and heat from friction. Stronger alternatives will be looked at and considered in the case that the durability is not enough.

3. Preparation for Manufacturing

The 3D printing files will be prepared for the helical gears and shaft supports, as well as laser cut files for the acrylic gearbox walls.

4. Structural Testing

Shaft stress analysis will be conducted for bending, torsion and fatigue life. The bearing load will be tested as well against the calculated radial and axial forces, in addition to evaluating gearbox efficiency.

5. Final Report and Presentation

Prepare all the final drawings and justification for each design decision. Test prototypes and evaluate outcomes, design improvements as well as cost analysis.

Contributions Sheet

There have been a number of instances where particular group members have not engaged in the project to an adequate level (sometimes making no contribution).

To take account of this, each group is required to list the percentage contribution of each group member (totaling 100%).

Each group member must sign this sheet in acknowledgment that they accept the contributions listed against them.

In the case where there are differences in the level of contribution by group members, the final individual group mark will be adjusted up or down.

Group number/name:

Group Member Name	**Percentage Contribution**
John [redacted]	25%
Jude [redacted]	25%
Noor [redacted]	25%
Shaik Ikram Jeelani	25%